```
import numpy as np
import pandas as pd
```

```
!pip install apyori
```

```
Requirement already satisfied: apyori in /usr/local/lib/python3.12/dist-packages (1.1.2)
```

header=none means first raw not behave for column

```
df=pd.read_csv("/content/market_basket - market_basket1.csv",header=None)
```

```
df.head(5)
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
0	shrimp	almonds	avocado	vegetables mix	green grapes	whole weat flour	yams	cottage cheese	energy drink	tomato juice	low fat yogurt	green tea	honey	salad	mineral water	salmon	antioxydant juice	frozen smoothie	spinach	olive oil
1	burgers	meatballs	eggs	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

```
df.shape
```

```
(7501, 20)
```

```
# data preprocessing
transactions=[]
for i in range(0,7501):
  transactions.append([str(df.values[i,j]) for j in range(0,20)])
```

```
# training the dataset
from apyori import apriori
rules=apriori(transactions=transactions,
              min_support=0.003,
              min_confidence=0.2,
              min_lift=3,
              min_length=2,)

a=list(rules)
result=[list(a[i][0]) for i in range(0,len(a))]
```

```
a
```

```
[RelationRecord(items=frozenset({'chicken', 'light cream'}), support=0.004532728969470737, ordered_statistics=[OrderedStatistic(items_base=frozenset({'light cream'}),
items_add=frozenset({'chicken'}), confidence=0.29059829059829057, lift=4.84395061728395)]),
 RelationRecord(items=frozenset({'mushroom cream sauce', 'escalope'}), support=0.005732568990801226, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'mushroom cream sauce'}), items_add=frozenset({'escalope'}), confidence=0.3006993006993007, lift=3.790832696715049)]),
 RelationRecord(items=frozenset({'pasta', 'escalope'}), support=0.005865884548726837, ordered_statistics=[OrderedStatistic(items_base=frozenset({'pasta'}),
items_add=frozenset({'escalope'}), confidence=0.3728813559322034, lift=4.700811850163794)]),
 RelationRecord(items=frozenset({'fromage blanc', 'honey'}), support=0.003332888948140248, ordered_statistics=[OrderedStatistic(items_base=frozenset({'fromage
blanc'}), items_add=frozenset({'honey'}), confidence=0.2450980392156863, lift=5.164270764485569)]),
 RelationRecord(items=frozenset({'herb & pepper', 'ground beef'}), support=0.015997866951073192, ordered_statistics=[OrderedStatistic(items_base=frozenset({'herb &
pepper'}), items_add=frozenset({'ground beef'}), confidence=0.3234501347708895, lift=3.2919938411349285)]),
 RelationRecord(items=frozenset({'tomato sauce', 'ground beef'}), support=0.005332622317024397, ordered_statistics=[OrderedStatistic(items_base=frozenset({'tomato
sauce'}), items_add=frozenset({'ground beef'}), confidence=0.3773584905660377, lift=3.840659481324083)]),
 RelationRecord(items=frozenset({'olive oil', 'light cream'}), support=0.003199573390214638, ordered_statistics=[OrderedStatistic(items_base=frozenset({'light
cream'}), items_add=frozenset({'olive oil'}), confidence=0.20512820512820515, lift=3.1147098515519573)]),
 RelationRecord(items=frozenset({'olive oil', 'whole wheat pasta'}), support=0.007998933475536596, ordered_statistics=[OrderedStatistic(items_base=frozenset({'whole
wheat pasta'}), items_add=frozenset({'olive oil'}), confidence=0.2714932126696833, lift=4.122410097642296)]),
 RelationRecord(items=frozenset({'shrimp', 'pasta'}), support=0.005065991201173177, ordered_statistics=[OrderedStatistic(items_base=frozenset({'pasta'}),
items_add=frozenset({'shrimp'}), confidence=0.3220338983050847, lift=4.506672147735896)]),
 RelationRecord(items=frozenset({'milk', 'avocado', 'spaghetti'}), support=0.003332888948140248, ordered_statistics=[OrderedStatistic(items_base=frozenset({'avocado',
'spaghetti'}), items_add=frozenset({'milk'}), confidence=0.41666666666666663, lift=3.215449245541838)]),
 RelationRecord(items=frozenset({'burgers', 'cake', 'milk'}), support=0.0037328356219170776, ordered_statistics=[OrderedStatistic(items_base=frozenset({'milk',
'cake'}), items_add=frozenset({'burgers'}), confidence=0.27999999999999997, lift=3.211437308868501)]),
 RelationRecord(items=frozenset({'burgers', 'chocolate', 'turkey'}), support=0.0030662578322890282, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chocolate', 'turkey'}), items_add=frozenset({'burgers'}), confidence=0.27058823529411763, lift=3.1034898363014927)]),
 RelationRecord(items=frozenset({'burgers', 'milk', 'turkey'}), support=0.003199573390214638, ordered_statistics=[OrderedStatistic(items_base=frozenset({'milk',
'turkey'}), items_add=frozenset({'burgers'}), confidence=0.2823529411764706, lift=3.2384241770102533)]),
 RelationRecord(items=frozenset({'cake', 'tomatoes', 'frozen vegetables'}), support=0.0030662578322890282, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'cake', 'frozen vegetables'}), items_add=frozenset({'tomatoes'}), confidence=0.2987012987012987, lift=4.367560314928736),
OrderedStatistic(items_base=frozenset({'cake', 'tomatoes'}), items_add=frozenset({'frozen vegetables'}), confidence=0.3650793650793650 6, lift=3.8300144300144296)]),
 RelationRecord(items=frozenset({'spaghetti', 'cereals', 'ground beef'}), support=0.0030662578322890282, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'cereals', 'ground beef'}), items_add=frozenset({'spaghetti'}), confidence=0.6764705882352942, lift=3.8853031258445188),
OrderedStatistic(items_base=frozenset({'spaghetti', 'cereals'}), items_add=frozenset({'ground beef'}), confidence=0.45999999999999996, lift=4.681763907734057)]),
 RelationRecord(items=frozenset({'chicken', 'milk', 'ground beef'}), support=0.0038661511798426876, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chicken', 'ground beef'}), items_add=frozenset({'milk'}), confidence=0.40845070422535207, lift=3.152046020981858)]),
 RelationRecord(items=frozenset({'chicken', 'nan', 'light cream'}), support=0.004532728969470737, ordered_statistics=[OrderedStatistic(items_base=frozenset({'light
cream'}), items_add=frozenset({'chicken', 'nan'}), confidence=0.29059829059829057, lift=4.84395061728395), OrderedStatistic(items_base=frozenset({'nan', 'light
cream'}), items_add=frozenset({'chicken'}), confidence=0.29059829059829057, lift=4.84395061728395)]),
 RelationRecord(items=frozenset({'chicken', 'olive oil', 'milk'}), support=0.0035995200639914677, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chicken', 'milk'}), items_add=frozenset({'olive oil'}), confidence=0.24324324324324323, lift=3.693456614509246),
OrderedStatistic(items_base=frozenset({'chicken', 'olive oil'}), items_add=frozenset({'milk'}), confidence=0.5, lift=3.858539094650206),
OrderedStatistic(items_base=frozenset({'milk', 'olive oil'}), items_add=frozenset({'chicken'}), confidence=0.2109375, lift=3.51609375)]),
 RelationRecord(items=frozenset({'chicken', 'olive oil', 'spaghetti'}), support=0.0034662045060658577, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chicken', 'spaghetti'}), items_add=frozenset({'olive oil'}), confidence=0.20155038759689922, lift=3.0603835169318647)]),
 RelationRecord(items=frozenset({'shrimp', 'chocolate', 'frozen vegetables'}), support=0.005332622317024397, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chocolate', 'frozen vegetables'}), items_add=frozenset({'shrimp'}), confidence=0.23255813953488375, lift=3.2545123221103784),
OrderedStatistic(items_base=frozenset({'shrimp', 'chocolate'}), items_add=frozenset({'frozen vegetables'}), confidence=0.29629629629629634,
lift=3.1084175084175087)]),
 RelationRecord(items=frozenset({'chocolate', 'herb & pepper', 'ground beef'}), support=0.003999466737768298, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'chocolate', 'herb & pepper'}), items_add=frozenset({'ground beef'}), confidence=0.4411764705882354,
lift=4.4901827759597746)]),
 RelationRecord(items=frozenset({'milk', 'chocolate', 'soup'}), support=0.003999466737768298, ordered_statistics=[OrderedStatistic(items_base=frozenset({'chocolate',
'soup'}), items_add=frozenset({'milk'}), confidence=0.3947368421052632, lift=3.0462150747238472)]),
 RelationRecord(items=frozenset({'cooking oil', 'spaghetti', 'ground beef'}), support=0.004799360085321957, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'cooking oil', 'ground beef'}), items_add=frozenset({'spaghetti'}), confidence=0.5714285714285714, lift=3.2819951870487856),
OrderedStatistic(items_base=frozenset({'cooking oil', 'spaghetti'}), items_add=frozenset({'ground beef'}), confidence=0.3025210084033613, lift=3.0789824749438446)],
```

```
 RelationRecord(items=frozenset({'herb & pepper', 'eggs', 'ground beef'}), support=0.0041327822956939075, ordered_statistics=
[OrderedStatistic(items_base=frozenset({'eggs', 'ground beef'}), items_add=frozenset({'herb & pepper'}), confidence=0.2066666666666667, lift=4.178454627133872),
```

```
result
```

```
[['chicken', 'light cream'],
 ['mushroom cream sauce', 'escalope'],
 ['pasta', 'escalope'],
 ['fromage blanc', 'honey'],
 ['herb & pepper', 'ground beef'],
 ['tomato sauce', 'ground beef'],
 ['olive oil', 'light cream'],
 ['olive oil', 'whole wheat pasta'],
 ['shrimp', 'pasta'],
 ['milk', 'avocado', 'spaghetti'],
 ['burgers', 'cake', 'milk'],
 ['burgers', 'chocolate', 'turkey'],
 ['burgers', 'milk', 'turkey'],
 ['cake', 'tomatoes', 'frozen vegetables'],
 ['spaghetti', 'cereals', 'ground beef'],
 ['chicken', 'milk', 'ground beef'],
 ['chicken', 'nan', 'light cream'],
 ['chicken', 'olive oil', 'milk'],
 ['chicken', 'olive oil', 'spaghetti'],
 ['shrimp', 'chocolate', 'frozen vegetables'],
 ['chocolate', 'herb & pepper', 'ground beef'],
 ['milk', 'chocolate', 'soup'],
 ['cooking oil', 'spaghetti', 'ground beef'],
 ['herb & pepper', 'eggs', 'ground beef'],
 ['eggs', 'spaghetti', 'red wine'],
 ['nan', 'mushroom cream sauce', 'escalope'],
 ['pasta', 'nan', 'escalope'],
 ['french fries', 'herb & pepper', 'ground beef'],
 ['nan', 'fromage blanc', 'honey'],
 ['green tea', 'tomatoes', 'frozen vegetables'],
 ['ground beef', 'spaghetti', 'frozen vegetables'],
 ['milk', 'olive oil', 'frozen vegetables'],
 ['milk', 'soup', 'frozen vegetables'],
 ['milk', 'tomatoes', 'frozen vegetables'],
 ['mineral water', 'shrimp', 'frozen vegetables'],
 ['olive oil', 'spaghetti', 'frozen vegetables'],
 ['shrimp', 'spaghetti', 'frozen vegetables'],
 ['shrimp', 'tomatoes', 'frozen vegetables'],
 ['tomatoes', 'spaghetti', 'frozen vegetables'],
 ['ground beef', 'spaghetti', 'grated cheese'],
 ['green tea', 'tomatoes', 'ground beef'],
 ['milk', 'herb & pepper', 'ground beef'],
 ['mineral water', 'herb & pepper', 'ground beef'],
 ['nan', 'herb & pepper', 'ground beef'],
 ['herb & pepper', 'spaghetti', 'ground beef'],
 ['milk', 'olive oil', 'ground beef'],
 ['milk', 'soup', 'ground beef'],
 ['nan', 'tomato sauce', 'ground beef'],
 ['pepper', 'spaghetti', 'ground beef'],
 ['shrimp', 'spaghetti', 'ground beef'],
```

```
 ['tomato sauce', 'spaghetti', 'ground beef'],
 ['mineral water', 'light cream', 'spaghetti'],
 ['olive oil', 'nan', 'light cream'],
 ['milk', 'olive oil', 'shrimp'],
 ['milk', 'olive oil', 'soup'],
 ['milk', 'olive oil', 'spaghetti'],
 ['milk', 'soup', 'tomatoes'],
 ['milk', 'whole wheat pasta', 'spaghetti'],
```

```
import pandas as pd
import numpy as np
```

```
df=pd.read_csv("/content/Social_Network_Ads - Social_Network_Ads.csv",header=None)
```

```
df.head(2)
```

	0	1	2	3	4
0	User ID	Gender	Age	EstimatedSalary	Purchased
1	15624510	Male	19	19000	0

```
df.size
```

```
2005
```

```
df.size
```

```
2005
```

```
# data preprocessing
transactions=[]
for i in range(1,df.shape[0]): # Start from 1 to skip header, iterate up to the last row
  transactions.append([str(df.values[i,j]) for j in range(0,df.shape[1])]) # Iterate over all actual columns
```

```
!pip install apyori
# training the dataset
from apyori import apriori
rules=apriori(transactions=transactions,
              min_support=0.003,
              min_confidence=0.2,
              min_lift=3,
              min_length=2,)

a=list(rules)
result=[list(a[i][0]) for i in range(0,len(a))]
```

```
Collecting apyori
  Downloading apyori-1.1.2.tar.gz (8.6 kB)
  Preparing metadata (setup.py) ... done
Building wheels for collected packages: apyori
  Building wheel for apyori (setup.py) ... done
  Created wheel for apyori: filename=apyori-1.1.2-py3-none-any.whl size=5954 sha256=1d8dfe18c647fea8e1f25351203458961e4d32df0547150e37b137888d65fd7c
  Stored in directory: /root/.cache/pip/wheels/7f/49/e3/42c73b19a264de37129fadaa0c52f26cf50e87de08fb9804af
Successfully built apyori
Installing collected packages: apyori
Successfully installed apyori-1.1.2
```

```
a
```

```
[RelationRecord(items=frozenset({'117000', '32'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'117000'}), items_add=frozenset({'32'}),
confidence=0.6666666666666667, lift=29.629629629629633), OrderedStatistic(items_base=frozenset({'32'}), items_add=frozenset({'117000'}),
confidence=0.22222222222222224, lift=29.629629629629633)]),
 RelationRecord(items=frozenset({'39', '134000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'134000'}), items_add=frozenset({'39'}),
confidence=0.5, lift=13.333333333333334)]),
 RelationRecord(items=frozenset({'26', '15000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'15000'}), items_add=frozenset({'26'}),
confidence=0.5, lift=12.5)]),
 RelationRecord(items=frozenset({'32', '18000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'18000'}), items_add=frozenset({'32'}),
confidence=0.5, lift=22.22222222222222), OrderedStatistic(items_base=frozenset({'32'}), items_add=frozenset({'18000'}), confidence=0.22222222222222224,
lift=22.222222222222225)]),
 RelationRecord(items=frozenset({'82000', '20'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'20'}), items_add=frozenset({'82000'}),
confidence=0.2857142857142857, lift=19.047619047619047), OrderedStatistic(items_base=frozenset({'82000'}), items_add=frozenset({'20'}),
confidence=0.33333333333333337, lift=19.047619047619047)]),
 RelationRecord(items=frozenset({'45', '22000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'22000'}), items_add=frozenset({'45'}),
confidence=0.39999999999999997, lift=22.857142857142854), OrderedStatistic(items_base=frozenset({'45'}), items_add=frozenset({'22000'}),
confidence=0.2857142857142857, lift=22.857142857142854)]),
 RelationRecord(items=frozenset({'24', '55000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'24'}), items_add=frozenset({'55000'}),
confidence=0.22222222222222224, lift=12.698412698412698), OrderedStatistic(items_base=frozenset({'55000'}), items_add=frozenset({'24'}),
confidence=0.2857142857142857, lift=12.698412698412698)]),
 RelationRecord(items=frozenset({'27', '58000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'58000'}), items_add=frozenset({'27'}),
confidence=0.39999999999999997, lift=12.307692307692307)]),
 RelationRecord(items=frozenset({'59000', '28'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'59000'}), items_add=frozenset({'28'}),
confidence=0.25, lift=8.333333333333334)]),
 RelationRecord(items=frozenset({'49', '28000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'28000'}), items_add=frozenset({'49'}),
confidence=0.33333333333333337, lift=13.333333333333334)]),
 RelationRecord(items=frozenset({'29', '43000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'43000'}), items_add=frozenset({'29'}),
confidence=0.2857142857142857, lift=11.428571428571427)]),
 RelationRecord(items=frozenset({'83000', '29'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'83000'}), items_add=frozenset({'29'}),
confidence=0.5, lift=20.0)]),
 RelationRecord(items=frozenset({'48', '33000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'33000'}), items_add=frozenset({'48'}),
confidence=0.33333333333333337, lift=9.523809523809524)]),
 RelationRecord(items=frozenset({'34', '43000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'34'}), items_add=frozenset({'43000'}),
confidence=0.33333333333333337, lift=19.047619047619047), OrderedStatistic(items_base=frozenset({'43000'}), items_add=frozenset({'34'}),
confidence=0.2857142857142857, lift=19.047619047619047)]),
 RelationRecord(items=frozenset({'35', '50000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'50000'}), items_add=frozenset({'35'}),
confidence=0.33333333333333337, lift=4.166666666666667)]),
 RelationRecord(items=frozenset({'38', '50000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'50000'}), items_add=frozenset({'38'}),
confidence=0.33333333333333337, lift=10.256410256410257)]),
 RelationRecord(items=frozenset({'38', '61000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'61000'}), items_add=frozenset({'38'}),
confidence=0.33333333333333337, lift=10.256410256410257)]),
 RelationRecord(items=frozenset({'38', '71000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'71000'}), items_add=frozenset({'38'}),
```

```
confidence=0.22222222222222224, lift=6.837606837606838)]),
 RelationRecord(items=frozenset({'39', '42000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'42000'}), items_add=frozenset({'39'}),
confidence=0.39999999999999997, lift=10.666666666666666)]),
 RelationRecord(items=frozenset({'39', '71000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'71000'}), items_add=frozenset({'39'}),
confidence=0.22222222222222224, lift=5.9259259259259265)]),
 RelationRecord(items=frozenset({'40', '57000'}), support=0.0075, ordered_statistics=[OrderedStatistic(items_base=frozenset({'40'}), items_add=frozenset({'57000'}),
confidence=0.2, lift=13.333333333333334), OrderedStatistic(items_base=frozenset({'57000'}), items_add=frozenset({'40'}), confidence=0.5, lift=13.333333333333334)]),
 RelationRecord(items=frozenset({'40', '75000'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'75000'}), items_add=frozenset({'40'}),
confidence=0.22222222222222224, lift=5.9259259259259265)]),
 RelationRecord(items=frozenset({'52000', '41'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'52000'}), items_add=frozenset({'41'}),
confidence=0.33333333333333337, lift=8.333333333333334)]),
 RelationRecord(items=frozenset({'72000', '41'}), support=0.01, ordered_statistics=[OrderedStatistic(items_base=frozenset({'41'}), items_add=frozenset({'72000'}),
confidence=0.25, lift=8.333333333333334), OrderedStatistic(items_base=frozenset({'72000'}), items_add=frozenset({'41'}), confidence=0.33333333333333337,
lift=8.333333333333334)]),
 RelationRecord(items=frozenset({'54000', '42'}), support=0.005, ordered_statistics=[OrderedStatistic(items_base=frozenset({'54000'}), items_add=frozenset({'42'}),
confidence=0.5, lift=12.5)]),
```

```
result
```

```
[['117000', '32'],
 ['39', '134000'],
 ['26', '15000'],
 ['32', '18000'],
 ['82000', '20'],
 ['45', '22000'],
 ['24', '55000'],
 ['27', '58000'],
 ['59000', '28'],
 ['49', '28000'],
 ['29', '43000'],
 ['83000', '29'],
 ['48', '33000'],
 ['34', '43000'],
 ['35', '50000'],
 ['38', '50000'],
 ['38', '61000'],
 ['38', '71000'],
 ['39', '42000'],
 ['39', '71000'],
 ['40', '57000'],
 ['40', '75000'],
 ['52000', '41'],
 ['72000', '41'],
 ['54000', '42'],
 ['65000', '42'],
 ['60', '42000'],
 ['26', '15000', '0'],
 ['32', '18000', '0'],
 ['Male', '18000', '0'],
 ['Male', '19000', '0'],
 ['82000', '20', '0'],
 ['24', '55000', '0'],
 ['Male', '25', '0'],
 ['27', '0', '58000'],
 ['59000', '0', '28'],
```

```
    ['29', '43000', '0'],
    ['83000', '29', '0'],
    ['34', '43000', '0'],
    ['35', '0', '50000'],
    ['35', '75000', '0'],
    ['38', '0', '50000'],
    ['38', '61000', '0'],
    ['38', '71000', '0'],
    ['39', '42000', '0'],
    ['39', '71000', '0'],
    ['40', '57000', '0'],
    ['40', '75000', '0'],
    ['52000', '41', '0'],
    ['72000', '41', '0'],
    ['54000', '42', '0'],
    ['42', '65000', '0'],
    ['44000', 'Female', '0'],
    ['62000', 'Female', '0'],
    ['68000', 'Female', '0'],
    ['Male', '81000', '0'],
    ['104000', 'Female', '1'],
    ['107000', 'Male', '1'],
```